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Project Charter

RMG Manufacturing & Business Performance Dashboard

Document Version History

Version	Date	Author	Description
1.0	9-Aug-2026	Shahadat Hossain Sajim	Initial Draft
1.1	20-Aug-2026	Shahadat Hossain Sajim	Final Portfolio Version

Project Information

Field	Value
Project Name	RMG Manufacturing & Business Performance Dashboard
Project Type	Data Analytics & Business Intelligence
Industry	RMG Sector
Role	Data Analyst
Document Status	Final
Confidentiality	Portfolio Project
Prepared By	Shahadat Hossain Sajim
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1. Business Background

The Ready-Made Garments (RMG) industry involves several connected activities such as customer orders, production, shipment, and quality control. Monitoring these activities separately can make it difficult to get a clear picture of overall business performance.

This project focuses on analyzing RMG business and operational data to understand how orders, buyers, factories, production, shipments, and quality processes are performing. The analysis is presented through an interactive Power BI dashboard to make the information easier to explore and understand.

2. Problem Statement

The available data contains information from different areas of the business, but the raw data does not provide a clear and convenient way to monitor overall performance. A structured analytical solution is needed to bring these data sources together, identify performance gaps, and provide meaningful insights for business and operational analysis.

3. Business Objectives

The main objectives of this project are to:

- Understand overall order and business performance.
- Compare buyer and order performance.
- Monitor shipment performance and freight costs.
- Evaluate production performance across factories and production lines.
- Measure production efficiency and achievement.
- Identify important quality and process issues.
- Track finishing rejects, dyeing loss, reprocessing, and packing performance.
- Present the findings through an interactive and easy-to-use Power BI dashboard.

4. Project Scope (In Scope / Out of Scope)

In Scope

- Data cleaning and preparation
- Data modeling and relationship development
- KPI and DAX measure creation
- Order and buyer analysis
- Shipment performance analysis
- Factory and production analysis
- Production line analysis
- Quality and process analysis
- Interactive Power BI dashboard development

Out Of Scope

- Predictive forecasting
- Machine learning
- Real-time production monitoring
- Detailed profitability or accounting analysis
- Automated recommendations or corrective actions

5. Key Business Questions

The dashboard is designed to answer the following business questions:

Order & Buyer

- What is the total order value and number of orders?
- Which buyers generate the highest order value?
- How does order value change over time?

Shipment

- What percentage of shipments are on time?

- How do planned shipments compare with actual shipments?
- Which factories have higher freight costs?

Production

- Which factories have the highest production achievement?
- Which production lines contribute the most output?
- Which factories have higher production efficiency?

Quality & Process

- What is the overall finishing reject rate?
- Which factories have higher finishing reject quantities?
- What is the dyeing batch loss rate?
- Which factories have higher dyeing reprocessing rates?
- How do finished and packed quantities change over time?
- How does reject quantity vary by month?

6. Key Performance Indicators

The dashboard uses a set of KPIs to monitor business, production, shipment, and quality performance.

Executive Overview

- Total Order Value
- Total Orders
- Production Achievement %
- On-Time Shipment %

Order, Buyer & Shipment

- Average Order Value
- Total Freight Cost
- On-Time Shipment %

Production & Factory Performance

- Actual Production
- Production Efficiency %
- Production Achievement %

- Lost Time %

Quality & Process Performance

- Finishing Reject %
- Dyeing Batch Loss %
- Dyeing Reprocess %
- Finishing Packing Rate %

7. Data & Tools

Tool	Purpose
Excel	Used for initial data inspection and profiling.
PostgreSQL	Used for data validation, data analysis, SQL queries, aggregations, and business analysis.
Power BI	Used for data modeling, visualization, dashboard development, filtering, and interactive reporting.
DAX	Used to create KPIs and business measures in Power BI.

8. Dashboard structure

The final dashboard is organized into four pages, with each page focusing on a specific area of the business.

Page 1 — Executive Overview

Provides a high-level view of order value, orders, production achievement, shipment status, monthly order trends, and factory production.

Page 2 — Order, Buyer & Shipment

Focuses on buyer performance, order value and quantity, freight cost, and planned versus actual shipments.

Page 3 — Production & Factory Performance

Focuses on actual production, production efficiency, production achievement across factories, and production performance by line.

Page 4 — Quality & Process Performance

Focuses on finishing rejects, dyeing batch loss, dyeing reprocessing, packing performance, and monthly quality trends.

9. Project Limitations

The analysis has some limitations based on the available data:

- The dashboard is based on historical data and does not provide real-time monitoring.
- The available dataset may not contain every factor that affects production or quality performance.
- Detailed profitability analysis is not included because sufficient financial cost data is not available.
- The project focuses on performance analysis rather than predictive forecasting.
- The dashboard highlights performance gaps but not their exact root causes.

10. Final Project Outcome

The project delivers an interactive Power BI dashboard that provides a clear view of order, shipment, production, factory, and quality performance. It helps users identify performance gaps, compare key business areas, and support data-driven decision-making.